8.1.) 단순회귀모형 p.186 참고

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \ , \ i = 1, 2, \cdots, n$$

을 적합시키기 위하여 n개의 데이터가 있다.

(1) 데이터에 절편만 있는 모형

$$y_i = \beta_0 + \epsilon_i \ , \ i = 1, \cdots, n$$

을 적합시킬 때 얻어지는 회귀제곱합을 $SS(b_0)$로 표시한다. 이 때

$$SS(b_0) = n(\bar{y})^2 = (\text{수정항})$$

임을 증명하라.

(Solution)

$\hat{\beta}_0 = \bar{y}$

절편만 있는 모형의 회귀제곱합 $SS(b_0)$는 원점(0)을 기준으로 한 적합값의 제곱합으로 정의됩니다.

$$SS(b_0) = \sum_{i=1}^{n} (\hat{y}_i)^2 = \sum_{1}^{n} \bar{y}^2 = n\bar{y}^2$$

* note:

1. 수정된 회귀제곱합 (Corrected SSR)

$\sum_{1}^{n} (\hat{y}_i - \bar{y})^2$는 "수정된 회귀제곱합 (Corrected Regression Sum of Squares)" 이다.

• 의미: 이 식은 '단순한 표본평균($\bar{y}$)을 사용했을 때보다, 설명변수(x)를 추가함으로써 모형이 얼마나 더 변동성을 잘 설명하게 되었는가'를 측정한다.

2. 문제의 정의: 비수정 회귀제곱합 (Uncorrected SSR), $SS(b_0)$

반면, 문제에서 정의하라고 한 $SS(b_0)$는 "비수정 (Uncorrected Regression Sum of Squares)"을 의미합니다.

• 의미: 데이터를 평균으로 중심화(centering)하지 않고, 원점(0)을 기준으로 계산한 제곱합입니다. 즉, β_0라는 파라미터 하나가 0에서부터 얼마나 데이터를 설명하고 있는지를 나타내는 '절편에 의한 회귀제곱합'입니다.

• 공식: 비수정 회귀제곱합의 정의는 편차가 아닌 단순 적합값의 제곱합인 $\sum_{i=1}^{n} \hat{y}_i^2$ 입니다.

(2) 데이터에 절편이 없는 모형

$$y_i = \beta_1 x_i + \epsilon_i \ , \ i = 1, \cdots, n$$

을 적합시킬 때 얻어지는 회귀제곱합을 $SS(b_1)$으로 표시한다. $SS(b_1)$을 x_i와 y_i로 표현하여라.

$$L(\beta_1) = \sum_{i=1}^{n} (y_i - \beta_1 x_i)^2 \ , \ \frac{\partial}{\partial \beta_1} L(\beta_1) = \sum_{1}^{n} -2x_i(y_i - \beta_1 x_i)$$

$$\left.\frac{\partial}{\partial \beta_1} L(\beta_1)\right|_{\beta = \beta_1} = 0 \ \therefore \hat{\beta}_1 = \frac{\overline{xy}}{\overline{x^2}} \ , \ \overline{xy} = \frac{1}{n}\sum_{1}^{n} x_i y_i \ , \ \overline{x^2} = \frac{1}{n}\sum_{1}^{n} x_i^2$$

$$SS(b_1) = \sum_{1}^{n} (\hat{y}_i)^2 = \sum_{1}^{n} \{x_i \hat{\beta}_1\}^2 = \hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2 = \left(\frac{\overline{xy}}{\overline{x^2}}\right)^2 \cdot \sum x_i^2 = \frac{(\sum x_i y_i)^2}{\sum x_i^2}$$

(3) 단순회귀모형을 적합시켰을 때 얻어지는 수정항을 빼지 않은 회귀제곱합, $SS(b_0, b_1)$의 표현식을 구하여라. 어떤 경우에

$$SS(b_0, b_1) = SS(b_0) + SS(b_1)$$

이 성립하는가? 이를 증명하라.

$$L(\beta_0, \beta_1) = \sum_{1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 \ , \ \beta_0 = \bar{y} - \hat{\beta}_1 \bar{x} \ \text{and} \ \beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$SS(b_0, b_1) = \sum_{1}^{n} (\hat{y}_i)^2 = \sum_{1}^{n} \hat{y}_i(\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 \sum_{1}^{n} \hat{y}_i + \hat{\beta}_1 \sum_{1}^{n} x_i \hat{y}_i = \hat{\beta}_0 \sum_{1}^{n} y_i + \hat{\beta}_1 \sum_{1}^{n} x_i y_i$$

$$SS(b_0, b_1) = SS(b_0) + SS(b_1 | b_0) = n(\bar{y})^2 + \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^{n} (\hat{\beta}_0 - \hat{\beta}_1 x_i - \bar{y})^2$$

$$SS(b_1 | b_0) = \sum_{i=1}^{n} (\hat{\beta}_0 - \hat{\beta}_1 x_i - \bar{y})^2 = \sum_{i=1}^{n} (\bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i - \bar{y})^2 = \sum_{1}^{n} (x_i - \bar{x})^2 \hat{\beta}_1$$

$$SS(b_0, b_1) = SS(b_0) + SS(b_1) \Leftrightarrow SS(b_1 | b_0) = SS(b_1)$$

$$\Rightarrow \sum_{1}^{n} (x_i - \bar{x})^2 \cdot \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \sum(x_i y_i)$$

$$\Rightarrow \sum(x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i$$

$$\Rightarrow \sum x_i = 0$$

(Approach #2)

$$SS(b_0, b_1) = \sum_{i=1}^{n} (\hat{y}_i)^2 = \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{n} (\bar{y} + \hat{\beta}_1 (x_i - \bar{x}))^2$$

$$= n(\bar{y})^2 + \hat{\beta}_1^2 \sum (x_i - \bar{x})^2$$

$$= n(\bar{y})^2 + S_{xy}^2 / S_{xx}$$

$$SS(b_0) + SS(b_1) = n(\bar{y})^2 + \frac{(\sum x_i y_i)^2}{\sum x_i^2}$$

$$\therefore SS(b_0, b_1) = SS(b_0) + SS(b_1)$$

$$\Leftrightarrow \frac{\{\sum(x_i - \bar{x})(y_i - \bar{y})^2\}^2}{\sum(x_i - \bar{x})^2} = \frac{(\sum x_i y_i)^2}{\sum x_i^2}$$

$$\Leftrightarrow \bar{x} = 0, \bar{y} = 0 \Leftrightarrow \sum x_i = \sum y_i = 0$$

8.2. 연습문제 6·2의 데이터에 대하여 다음 질문에 답하여라. 중회귀모형, $y_j = \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + \epsilon_j$, $\epsilon_j \sim N(0, \sigma^2)$이 적절하다고 가정한다.

(1) $SS(b_1)$, $SS(b_1|b_0)$와 $SS(b_1|b_0, b_2)$를 각각 구하여라.

(2) $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$
을 부분 F-검정으로 가설검정하라. 유의수준은 $\alpha = 0.05$

(3) x_1과 x_2 중 하나의 변수만을 선택하여 회귀식을 구한다면 어떤 변수를 택하겠는가? 어떤 기준으로 이 변수를 택하였는가?

(Solution)

(1) $y_j = \beta_1 x_{1j}$, $L(\beta_1) = \sum_i^n (y_i - x_{1i}\beta_1)^2$, $\partial L/\partial \beta_1 = \sum_i^n -2x_{1i}(y_i - \beta_1 x_{1i})$

$$\therefore \hat{\beta}_1 = \frac{\sum x_{1i} \cdot y_i}{\sum x_{1i}^2} = 0.568596\ldots$$

$$SS(b_1) = \sum_{i=1}^n (\hat{y}_i)^2 = \sum_{i=1}^n (x_{1i}\hat{\beta}_1)^2 = 102200.1524$$

$$SS(b_1|b_0) = \sum_{i=1}^n (x_{1i}\hat{\beta}_1 + \hat{\beta}_0 - \bar{y})^2 = \sum_i^n (x_{1i} - \bar{x})^2 \underline{\hat{\beta}_1} = \sum_i^n (x_{1i} - \bar{x})(y_i - \bar{y}) = 73$$

여기서 $\hat{\beta}_1$은 위의 $\hat{\beta}_1$과 다른 것이다. 적합시킨 모형이 다르기 때문

8.3. 회귀모형, $y_j = \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + \epsilon_j$, $j = 1, \cdots, n$ 에서 만약 $\sum_{j=1}^n x_{1j} x_{2j} = 0$, $\bar{x}_1 = 0$ 또는 $\bar{x}_2 = 0$ 이면 $SS(b_1|b_0) = SS(b_1|b_0, b_2)$임을 증명하라.

$$y = \mathbb{1}\beta_0 + X_1\beta_1 + X_2\beta_2 + \epsilon$$
$$SS(\beta_1, \beta_2|\beta_0) = y^T(\Pi_X - \Pi_{\mathbb{1}})y$$
$$SS(\beta_1|\beta_0, \beta_2) = y^T(\Pi_X - \Pi_{\mathbb{1}} - \Pi_{X_{2,\perp}})y$$
$$SS(\beta_1|\beta_0) = y^T(\Pi_{X_1} - \Pi_{\mathbb{1}})y$$

* $\Pi_X = X(X^TX)^{-1}X^T$, $\Pi_{X_{2,\perp}} = X_{2,\perp}(X_{2,\perp}^T X_{2,\perp})^{-1} X_{2,\perp}^T$, $X_{2,\perp} = X_2 - \Pi_{\mathbb{1}} X_2$
$\Pi_{X_{1,\perp}} = X_{1,\perp}(X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T$, $X_{1,\perp} = X_1 - \Pi_{\mathbb{1}} X_1$

Want to show (WTS) : $SS(\beta_1|\beta_0, \beta_2) = SS(\beta_1|\beta_0)$
$\Leftrightarrow y^T(\Pi_X - \Pi_{\mathbb{1}} - \Pi_{X_{2,\perp}})y = y^T(\Pi_{X_{1,\perp}})y$

If $\sum x_{1j} x_{2j} = 0 \Leftrightarrow C(X_1) \perp C(X_2)$, $C(X_i)$: column space of X_i
If $\bar{x}_1 = 0 \Leftrightarrow x_1^T \mathbb{1} = 0 \Leftrightarrow C(X_1) \perp C(\mathbb{1})$
If $\bar{x}_2 = 0 \Leftrightarrow x_2^T \mathbb{1} = 0 \Leftrightarrow C(X_2) \perp C(\mathbb{1})$

(case 1) : $\sum x_{1j} x_{2j} = 0$ & $\bar{x}_1 = 0$

LHS: $y^T(\Pi_X - \Pi_{\mathbb{1}} - \Pi_{X_{2,\perp}})y = y^T(\Pi_{\mathbb{1}} + \Pi_{X_{2,\perp}} + \Pi_{X_{1,\mathbb{1}}} - \Pi_{\mathbb{1}} - \Pi_{X_{2,\perp}})y$
$= y^T(\Pi_{X_{1,\mathbb{1}}})y$, $X_{1,\mathbb{1}} = X_1 - \Pi_{\mathbb{1}} X_1 - \Pi_{X_{2,\perp}}$
$= y^T(\Pi_{X_1})y \quad \because C(X_1) \perp C(X_2)$ & $C(X_1) \perp C(\mathbb{1})$

RHS: $y^T(\Pi_{X_{1,\perp}})y = y^T(\Pi_{X_1})y \quad \because C(X_1) \perp C(\mathbb{1})$

$\therefore$ LHS = RHS

(case 2) : $\sum x_{1j} x_{2j} = 0$ & $\bar{x}_2 = 0$

LHS : $y^T(\Pi_X - \Pi_{\mathbb{1}} - \Pi_{X_{2,\perp}})y = y^T(\Pi_X - \Pi_{\mathbb{1}} - \Pi_{X_2})y \quad \because C(X_2) \perp C(\mathbb{1})$
$= y^T(\Pi_{\mathbb{1}} + \Pi_{X_{2,\perp}} + \Pi_{X_{1,\mathbb{1}}} - \Pi_{\mathbb{1}} - \Pi_{X_2})y$
$= y^T(\Pi_{X_{1,\mathbb{1}}})y = y(\Pi_{X_{1,\perp}})y \quad \because C(X_1) \perp C(X_2)$
$=$ RHS

8.8.) 선형회귀모형 $y=X\beta+\epsilon$ 에서 결정계수 R^2은 y_j 나 x_{ij}의 단위가 바뀌어도 관계 없이 일정함을 증명하라. 즉 선형변환

$$z_j = c y_j,\ w_{ij} = c_i x_{ij},\ i=1,\cdots,k,\ j=1,2,\cdots,n$$

을 시킨 후의 R^2과 시키기 전의 R^2이 같음을 증명하라.

(Solution)

$$R^2 = \frac{SSR}{SST} = \frac{y^T(\Pi_X - \Pi_{\mathbb{1}})y}{y^T(I_n - \Pi_{\mathbb{1}})y} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$(R^2)' = \frac{SSR'}{SST'} = \frac{z^T(\Pi_W - \Pi_{\mathbb{1}})z}{z^T(I_n - \Pi_{\mathbb{1}})z} = \frac{\sum_{i=1}^n (\hat{z}_i - \bar{z})^2}{\sum_{i=1}^n (z_i - \bar{z})^2}$$

$$z = [z_1 \cdots z_n]^T = [cy_1 \cdots cy_n]^T \in \mathbb{R}^{n\times 1}$$
$$= cI_n y \quad (c\in\mathbb{R})$$

$$W = [w_{11}\ w_{21} \cdots w_{k1}] = [c_1 x_1 \cdots c_k x_k] \in \mathbb{R}^{n\times k}$$
$$= \underbrace{\underbrace{\mathrm{diag}(c_1,\cdots,c_n)}_{\in\mathbb{R}^{n\times n}}\underbrace{[x_1 \cdots x_k]}_{\in\mathbb{R}^{n\times k}}}_{\text{순서가 바뀌어야 함}} = c'X$$

XC 가 맞음

$$\frac{z^T(\Pi_W - \Pi_{\mathbb{1}})z}{z^T(I_n - \Pi_{\mathbb{1}})z} = \frac{(cy)^T(\Pi_X - \Pi_{\mathbb{1}})(cy)}{(cy)^T(I_n - \Pi_{\mathbb{1}})(cy)} = \frac{y^T(\Pi_X - \Pi_{\mathbb{1}})y}{y^T(I_n - \Pi_{\mathbb{1}})y}$$

Π_W : projection onto the column space of $\mathrm{diag}(c_1\cdots c_k)X$

$\Rightarrow$ projection onto the column space of X

$\Rightarrow \Pi_X$

$\therefore \Pi_W = \Pi_X$

[Textbook Approach]

선형변환 후 b의 최소제곱추정값을 $\hat{r}$이라 하자.

만약

$$C = \begin{bmatrix} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & c_k \end{bmatrix},\quad z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

이라 놓고, $W = XC$라 하면 다음이 성립한다.

$$\hat{r} = (W^TW)^{-1}W^Tz$$
$$= (C^TX^TXC)^{-1}C^TX^Tz$$
$$= C^{-1}(X^TX)^{-1}C^{-1}CX^Tz$$
$$= cC^{-1}b \quad (= cC^{-1}\hat{\beta})$$

$$\hat{z} = W\hat{r} = cWC^{-1}b$$
$$= cXCC^{-1}b$$
$$= cXb$$
$$= c\hat{y}$$

따라서 선형변환시킨 후의 R^2을 R_z^2 라고 하면

$$R_z^2 = \frac{\sum_i^n (\hat{z}_i - \bar{z})^2}{\sum_i^n (z_i - \bar{z})^2} = \frac{c^2\sum_i^n (\hat{y}_i - \bar{y})^2}{c^2\sum_i^n (y_i - \bar{y})^2} = R^2$$

8.9. 두 회귀선

$$y_{1j} = \beta_1 x_j + \epsilon_{1j},\quad y_{2j} = \beta_2 x_j + \epsilon_{2j},\quad j=1,\cdots,n$$

에서 가설 $H_0: \beta_1 = \beta_2$를 검정하기 위한 F-통계량을 구하라.

다음을 가정하라.

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix} \sim N(0, \sigma^2 I_n)$$

$$\frac{SSE(R) - SSE(F) / (k+1)}{SSE(F)/(n-k-1)} \sim F(k+1, n-k-1)$$

$$\text{Full Model}: \begin{cases} y_1 = X_1\beta_1 + \epsilon_1 \\ y_2 = X_2\beta_2 + \epsilon_2 \end{cases}$$

$$\text{Sub Model}: \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}\beta_0 + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix} \quad : y = X\beta + \epsilon$$

$$SSE(F) = y_1^T(\Pi_{X_1} - \Pi_{\mathbb{1}_{n_1}})y_1 + y_2^T(\Pi_{X_2} - \Pi_{\mathbb{1}_{n_2}})y_2$$
$$= \sum_{i=1}^n (\hat{y}_{1i} - \bar{y}_1)^2 + \sum_{i=1}^n (\hat{y}_{2i} - \bar{y}_2)^2$$

$$\hat{y}_{1i} = X_1(X_1^TX_1)^{-1}X_1^Ty_{1i},\quad \hat{y}_{2i} = X_2(X_2^TX_2)^{-1}X_2^Ty_{2i}$$
$$= X_1 b_1 \qquad\qquad = X_2 b_2$$

$$\bar{y}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} y_{1i},\quad \bar{y}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} y_{2i}$$

$$SSE(R) = y^T(\Pi_X - \Pi_{\mathbb{1}})y = \sum_{i=1}^{2}\sum_{j=1}^{n_i}(\hat{y}_{ij} - \bar{y})^2,\quad \bar{y} = \frac{1}{n}\sum_{i=1}^{2}\sum_{j=1}^{n_i} y_{ij},\quad n = n_1 + n_2$$

$$\therefore \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-p-1)} \sim F(p+1, n-p-1) \quad \text{under } H_0$$

[Textbook Approach]

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{1n} \\ y_{21} \\ y_{22} \\ \vdots \\ y_{2n} \end{bmatrix} = \begin{bmatrix} x_1 & 0 \\ x_2 & 0 \\ \vdots & \vdots \\ x_n & 0 \\ 0 & x_1 \\ 0 & x_2 \\ \vdots & \vdots \\ 0 & x_n \end{bmatrix}\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} + \epsilon$$

$$H_0: C\beta = 0 \Rightarrow C = [1\ -1]$$

$$F_0 = \frac{(C\beta)'[C(X'X)^{-1}C']^{-1}Cb}{MSE} = \frac{(b_1 - b_2)^2\sum x_i^2}{2\,MSE}$$

$$y = \begin{bmatrix} y_{11} \\ \vdots \\ y_{1n} \\ y_{21} \\ \vdots \\ y_{2n} \end{bmatrix}, \quad X = \begin{bmatrix} x_{11} & 0 \\ \vdots & \vdots \\ x_{1n} & 0 \\ 0 & x_{21} \\ \vdots & \vdots \\ 0 & x_{2n} \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \quad \epsilon = \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{1n} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{2n} \end{bmatrix}$$

$$y = X\beta, \quad F = \frac{(A\hat{\beta})^T (A(X^TX)^{-1}A^T)^{-1}(A\hat{\beta})/q}{SSE/(n-p-1)}$$

$A\beta = 0 \quad \therefore A = [1 \ \ -1], \ q = 1$

$$X^TX = \begin{bmatrix} \sum_{j=1}^{n} x_{1j}^2 & 0 \\ 0 & \sum_{j=1}^{n} x_{2j}^2 \end{bmatrix} \quad (X^TX)^{-1} = \begin{bmatrix} (\sum_{j=1}^{n} x_{1j}^2)^{-1} & 0 \\ 0 & (\sum_{j=1}^{n} x_{2j}^2)^{-1} \end{bmatrix}$$

$$A(X^TX)^{-1}A^T = \left\{\sum_{j=1}^{n} x_{1j}^2\right\}^{-1} + \left\{\sum_{j=1}^{n} x_{2j}^2\right\}^{-1}$$

$(A(X^TX)^{-1}A^T)^{-1} = ($ becomes overly complicated)

$$F = \frac{(A\hat{\beta})^T (A(X^TX)^{-1}A^T)^{-1}(A\hat{\beta})/1}{y^T(I_n - \Pi_X)y/(n-2)}, \quad \hat{\beta} = (X^TX)^{-1}X^Ty, \quad \Pi_X = X(X^TX)^{-1}X^T$$

$F \sim F(1, n-2)$